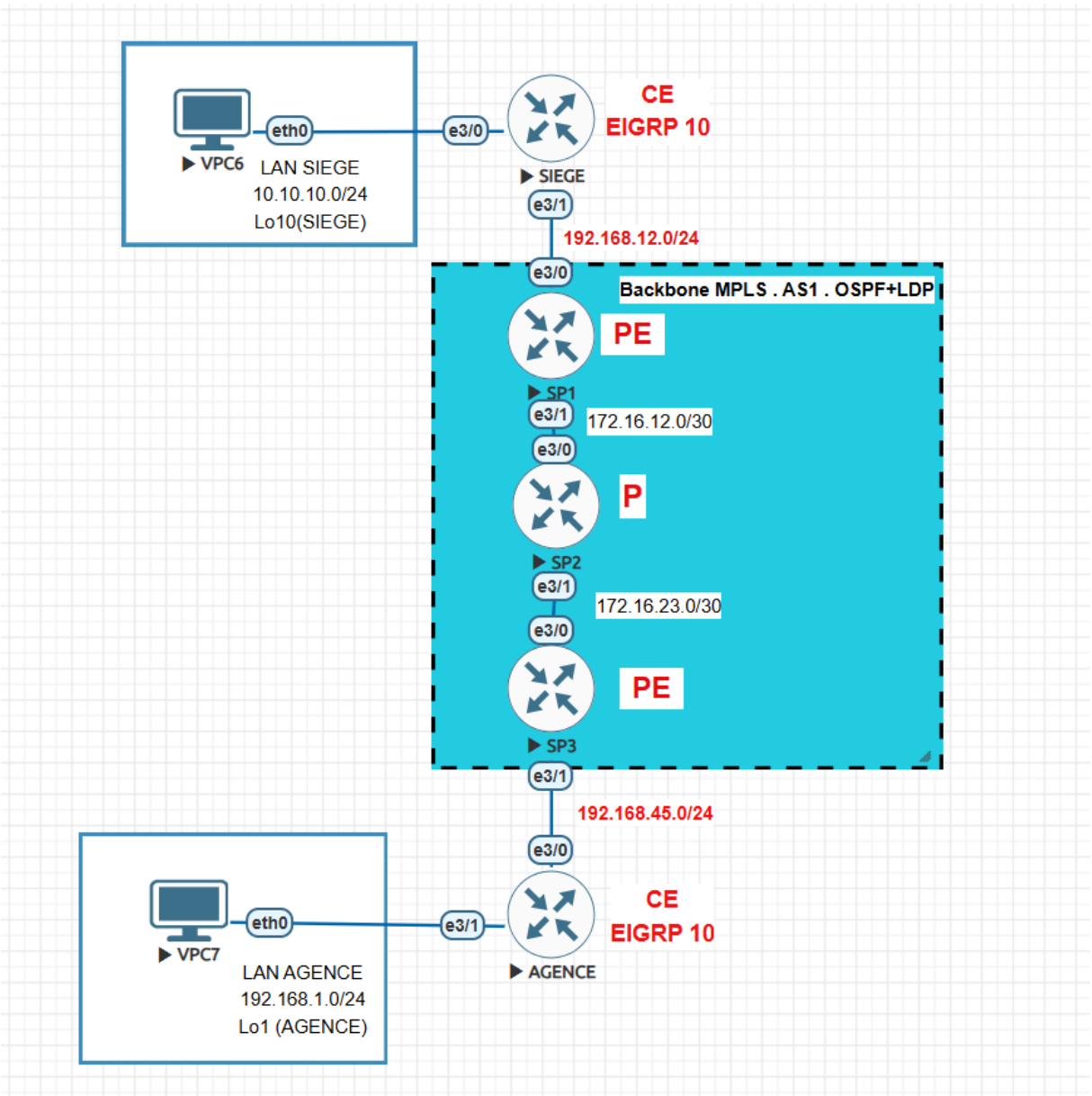
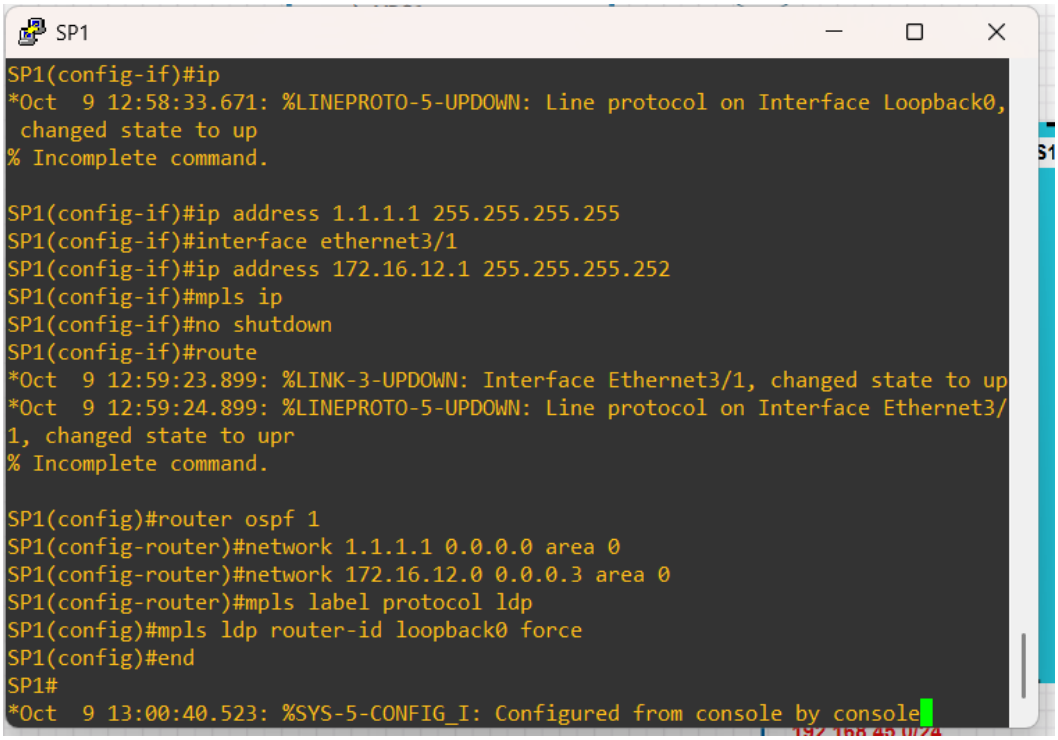
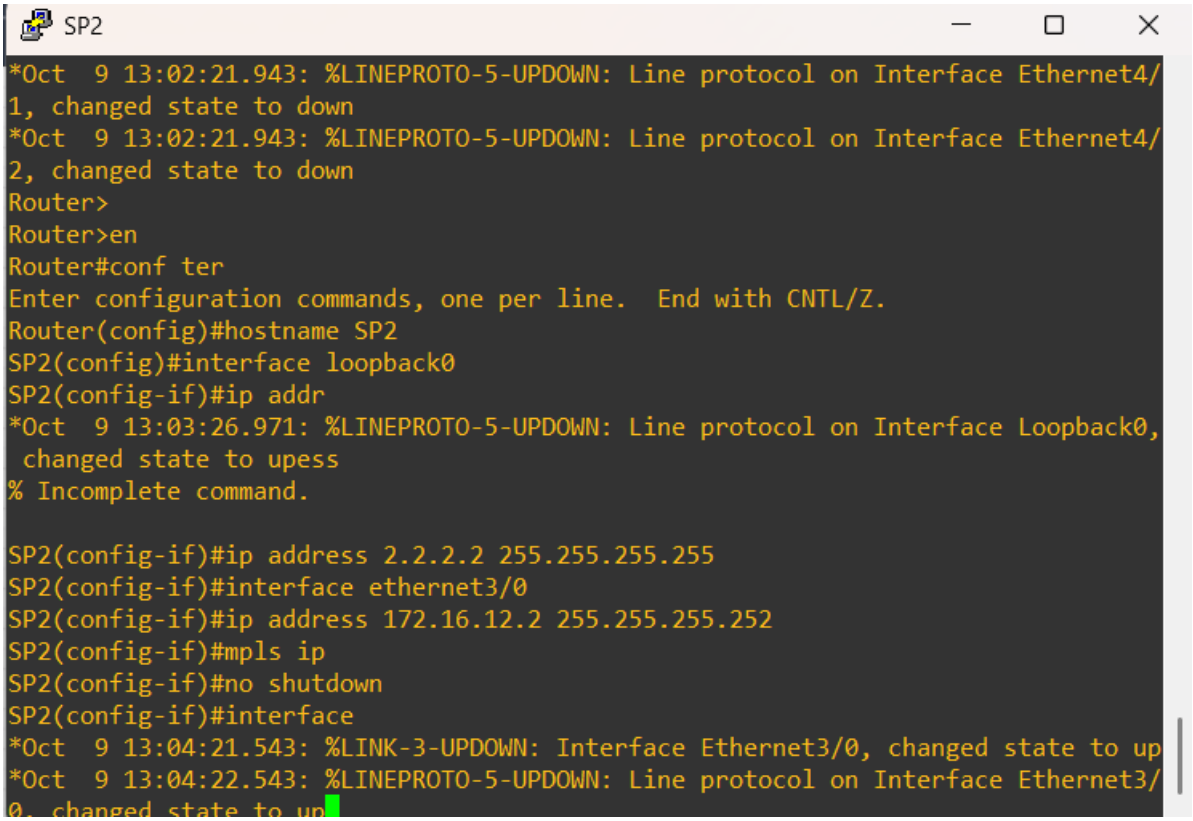
Compte-Rendu : TP2- VPN MPLS

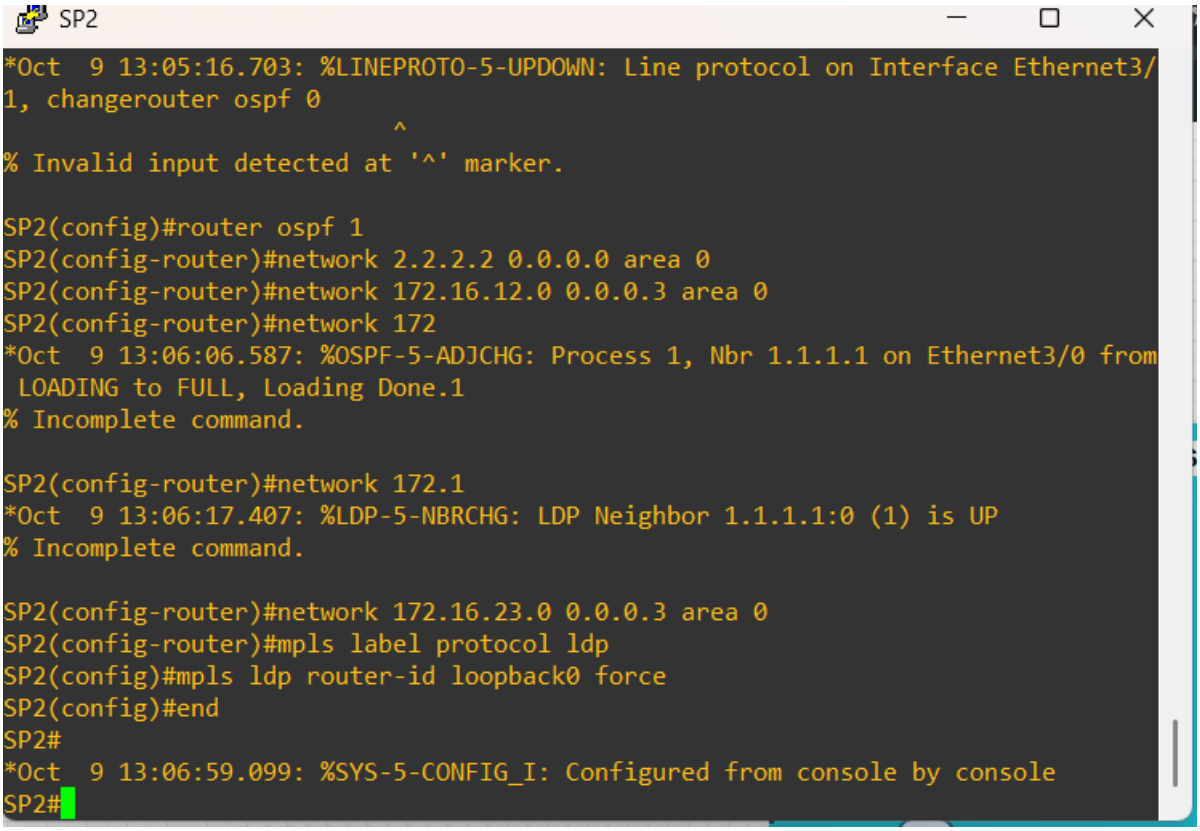
Malek Boughdiri RSI31

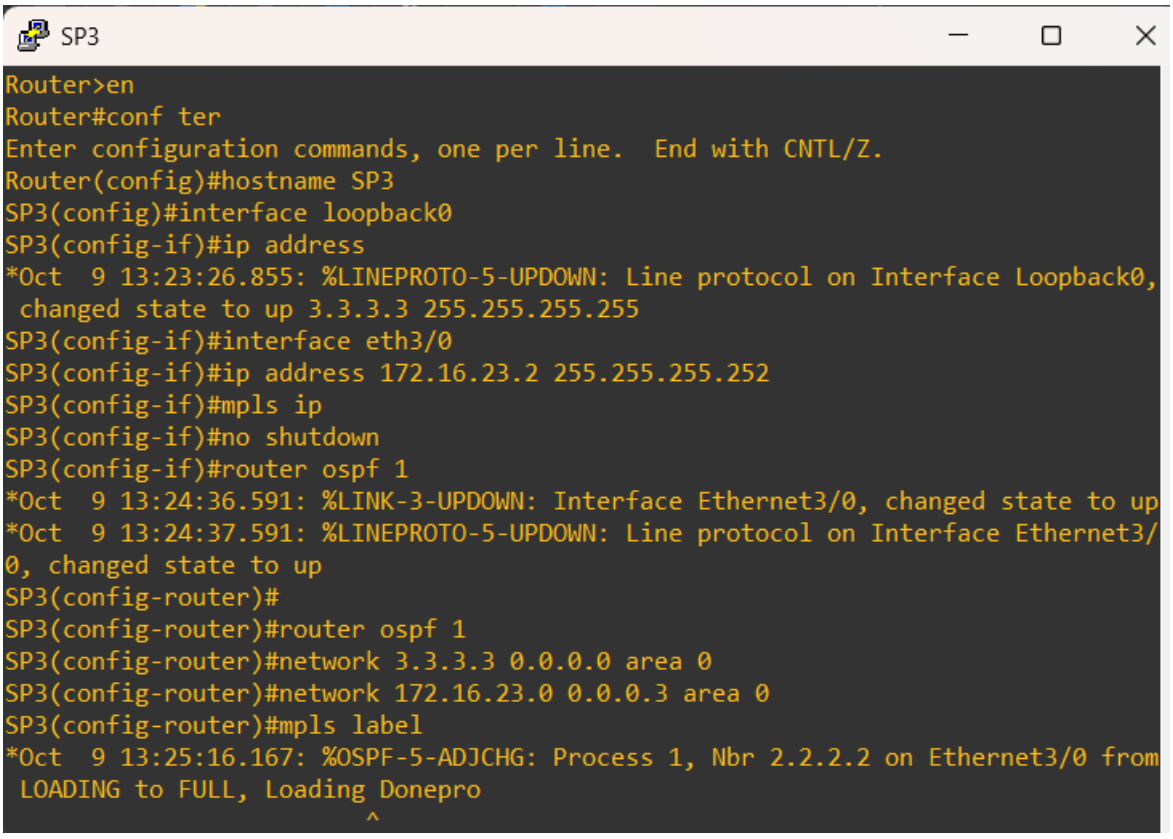


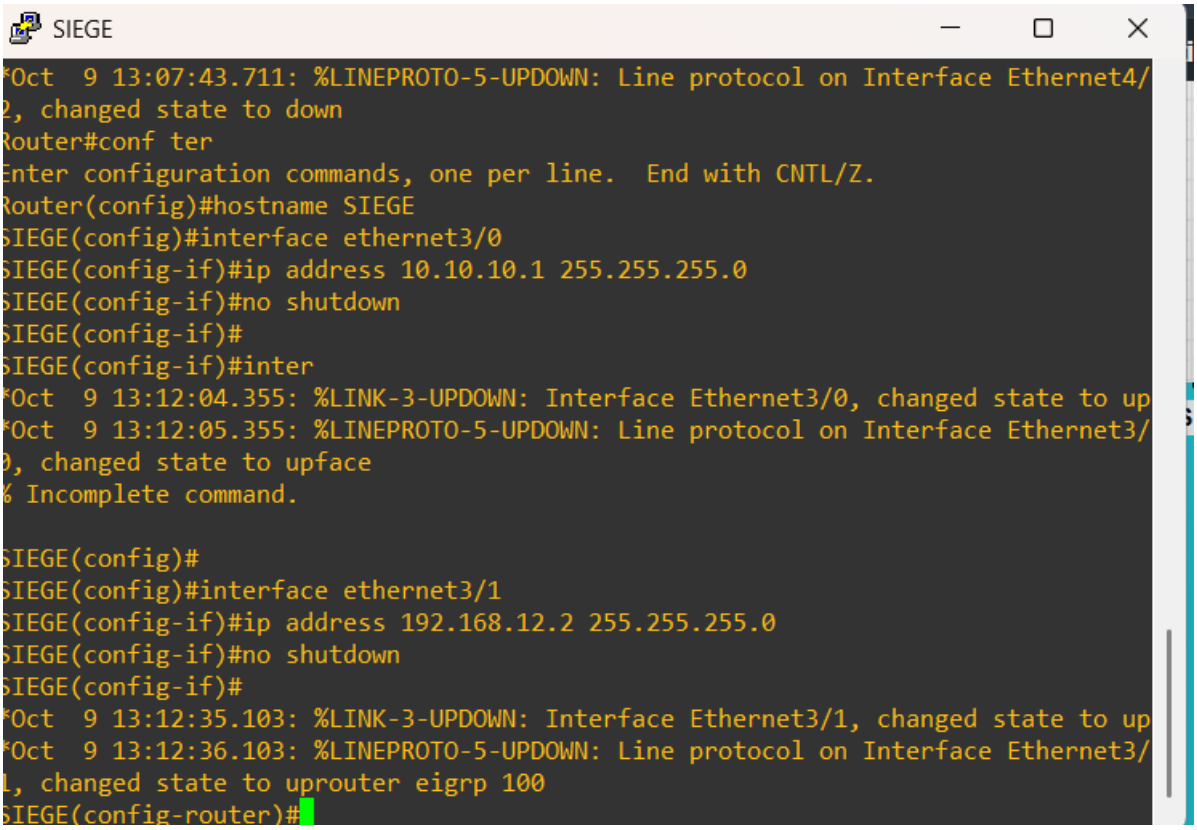
Étape 0 : préparation (backbone OSPF + LDP et réseau client) :

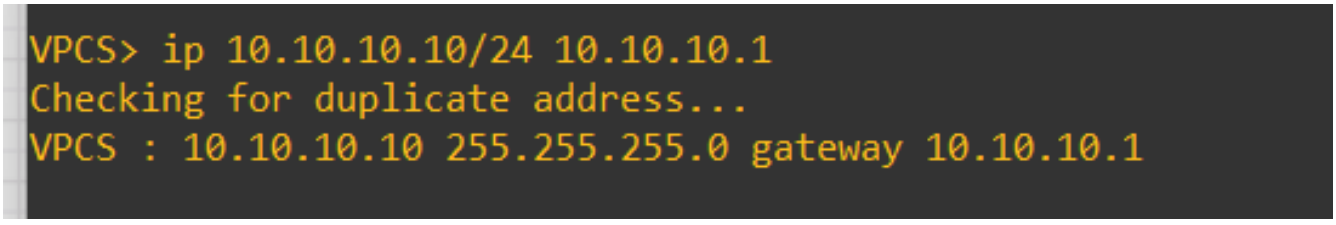


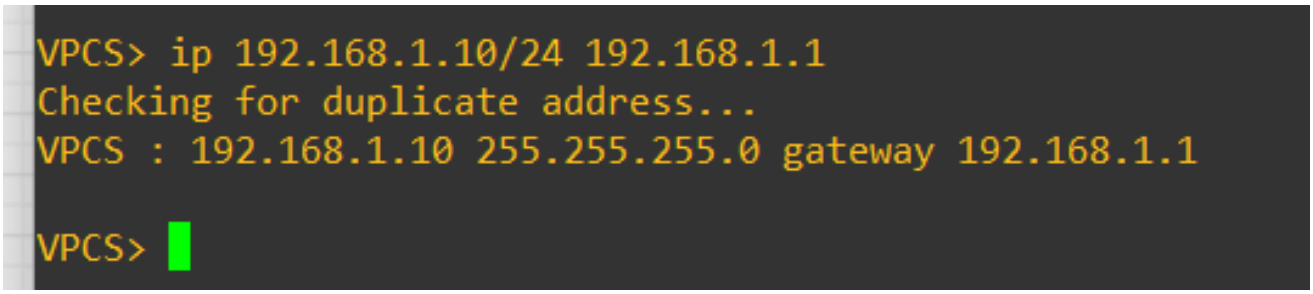



SP2
*Oct 9 13:05:16.703: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/
1, changerouter ospf 0
^
% Invalid input detected at '^' marker.
SP2(config)#router ospf 1
SP2(config-router)#network 2.2.2.2 0.0.0.0 area 0
SP2(config-router)#network 172.16.12.0 0.0.0.3 area 0
SP2(config-router)#network 172
*Oct 9 13:06:06.587: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Ethernet3/0 from
LOADING to FULL, Loading Done.1
% Incomplete command.
SP2(config-router)#network 172.1
*Oct 9 13:06:17.407: %LDP-5-NBRCHG: LDP Neighbor 1.1.1.1:0 (1) is UP
% Incomplete command.
SP2(config-router)#network 172.16.23.0 0.0.0.3 area 0
SP2(config-router)#mpls label protocol ldp
SP2(config)#mpls ldp router-id loopback0 force
SP2(config)#end
SP2#
*Oct 9 13:06:59.099: %SYS-5-CONFIG_I: Configured from console by console
SP2#


SP3
Router>en
Router#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname SP3
SP3(config)#interface loopback0
SP3(config-if)#ip address
*Oct 9 13:23:26.855: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
changed state to up 3.3.3.3 255.255.255.255
SP3(config-if)#interface eth3/0
SP3(config-if)#ip address 172.16.23.2 255.255.255.252
SP3(config-if)#mpls ip
SP3(config-if)#no shutdown
SP3(config-if)#router ospf 1
*Oct 9 13:24:36.591: %LINK-3-UPDOWN: Interface Ethernet3/0, changed state to up
*Oct 9 13:24:37.591: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/
0, changed state to up
SP3(config-router)#
SP3(config-router)#router ospf 1
SP3(config-router)#network 3.3.3.3 0.0.0.0 area 0
SP3(config-router)#network 172.16.23.0 0.0.0.3 area 0
SP3(config-router)#mpls label
*Oct 9 13:25:16.167: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet3/0 from
LOADING to FULL, Loading Donepro
^


SIEGE
*Oct 9 13:07:43.711: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet4/
2, changed state to down
Router#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname SIEGE
SIEGE(config)#interface ethernet3/0
SIEGE(config-if)#ip address 10.10.10.1 255.255.255.0
SIEGE(config-if)#no shutdown
SIEGE(config-if)#
SIEGE(config-if)#inter
*Oct 9 13:12:04.355: %LINK-3-UPDOWN: Interface Ethernet3/0, changed state to up
*Oct 9 13:12:05.355: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/
0, changed state to upface
% Incomplete command.
SIEGE(config)#
SIEGE(config)#interface ethernet3/1
SIEGE(config-if)#ip address 192.168.12.2 255.255.255.0
SIEGE(config-if)#no shutdown
SIEGE(config-if)#
*Oct 9 13:12:35.103: %LINK-3-UPDOWN: Interface Ethernet3/1, changed state to up
*Oct 9 13:12:36.103: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/
1, changed state to uprouter eigrp 100
SIEGE(config-router)#


VPCS> ip 10.10.10.10/24 10.10.10.1
Checking for duplicate address...
VPCS : 10.10.10.10 255.255.255.0 gateway 10.10.10.1


VPCS> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1
VPCS>

SIEGE

```
SIEGE(config-if)#no shutdown
SIEGE(config-if)#
SIEGE(config-if)#inter
*Oct  9 13:12:04.355: %LINK-3-UPDOWN: Interface Ethernet3/0, changed state to up
*Oct  9 13:12:05.355: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/
0, changed state to upface
% Incomplete command.

SIEGE(config)#
SIEGE(config)#interface ethernet3/1
SIEGE(config-if)#ip address 192.168.12.2 255.255.255.0
SIEGE(config-if)#no shutdown
SIEGE(config-if)#
*Oct  9 13:12:35.103: %LINK-3-UPDOWN: Interface Ethernet3/1, changed state to up
*Oct  9 13:12:36.103: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/
1, changed state to uprouter eigrp 100
SIEGE(config-router)#router eigrp 100
SIEGE(config-router)#network 10.10.10.0 0.0.0.255
SIEGE(config-router)#network 192.168.12.0  0.0.0.255
SIEGE(config-router)#no auto-summary
SIEGE(config-router)#end
SIEGE#
*Oct  9 13:13:44.827: %SYS-5-CONFIG_I: Configured from console by console
SIEGE#
```

AGENCE

```
0, changed state to down
*Oct  9 13:14:26.227: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet4/
1, changed state to down
*Oct  9 13:14:26.227: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet4/
2, changed state to down
Router>
Router>en
Router#en
Router#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname AGENCE
AGENCE(config)#interface eth3/0
AGENCE(config-if)#ip address 192.168.45.2 255.255.255.0
AGENCE(config-if)#no shutdown
AGENCE(config-if)#interface
*Oct  9 13:15:16.847: %LINK-3-UPDOWN: Interface Ethernet3/0, changed state to up
*Oct  9 13:15:17.847: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/
0, changed state to up
% Incomplete command.

AGENCE(config)#interface eth3/1
AGENCE(config-if)#ip address 192.168.1.1 255.255.255.0
AGENCE(config-if)#no shutdown
AGENCE(config-if)#router eig
```

AGENCE

```
AGENCE(config-if)#ip address 192.168.45.2 255.255.255.0
AGENCE(config-if)#no shutdown
AGENCE(config-if)#interface
*Oct  9 13:15:16.847: %LINK-3-UPDOWN: Interface Ethernet3/0, changed state to up
*Oct  9 13:15:17.847: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/
0, changed state to up
% Incomplete command.

AGENCE(config)#interface eth3/1
AGENCE(config-if)#ip address 192.168.1.1 255.255.255.0
AGENCE(config-if)#no shutdown
AGENCE(config-if)#router eig
*Oct  9 13:15:41.583: %LINK-3-UPDOWN: Interface Ethernet3/1, changed state to up
*Oct  9 13:15:42.583: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/
1, changed state to uprp
% Incomplete command.

AGENCE(config)#router eigrp 100
AGENCE(config-router)#network 192.168.45.0 0.0.0.255
AGENCE(config-router)#network 192.168.1.0 0.0.0.255
AGENCE(config-router)#no auto-summary
AGENCE(config-router)#end
AGENCE#
*Oct  9 13:16:36.075: %SYS-5-CONFIG_I: Configured from console by console
```

Vérifications du backbone :

```
SP1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/BDR        00:00:38    172.16.12.2     Ethernet3/1
SP1#
```

```
SP1#show mpls ldp neighbor
    Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 1.1.1.1:0
        TCP connection: 2.2.2.2.43339 - 1.1.1.1.646
        State: Oper; Msgs sent/rcvd: 36/36; Downstream
        Up time: 00:25:03
        LDP discovery sources:
          Ethernet3/1, Src IP addr: 172.16.12.2
        Addresses bound to peer LDP Ident:
          172.16.12.2     2.2.2.2         172.16.23.1
SP1#
```

```
SP1#ping 3.3.3.3 source loopback0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
Packet sent with a source address of 1.1.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/24/32 ms
```

Étape 1 : instance VRF sur SP1 et SP3

1. Créer la VRF

```
SP1(config)#ip vrf Client_A
```

```
SP3(config)#ip vrf Client_A
```

2. Configurer RD et RT (sur SP1 et SP3)

```
SP1(config)#ip vrf Client_A
SP1(config-vrf)#ip vrf Client_A
SP1(config-vrf)#rd 100:1
SP1(config-vrf)#route-target both 1:100
SP1(config-vrf)#exit
```

```
SP3(config)#ip vrf Client_A
SP3(config-vrf)#ip vrf Client_A
SP3(config-vrf)#rd 100:1
SP3(config-vrf)#route-target both 1:100
SP3(config-vrf)#exit
```

SP1 reçoit une route VPNv4 avec la Route Target 100:100. BGP va-t-il l'importer dans la VRF du client ?

Non. La VRF n'importe que les routes portant la RT 1:100 (route-target both 1:100 fait import et export). Une route portant 100:100 est ignorée.

3. Associer la VRF à l'interface face au CE

```
SP1(config)#interface eth3/0
SP1(config-if)#ip vrf forwarding Client_A
```

```
SP3(config)#interface eth3/1
SP3(config-if)#ip vrf forwarding Client_A
```

4. Ressaisir les adresses IP

```
SP1(config-if)#ip address 192.168.12.1 255.255.255.0
SP1(config-if)#no shutdown
```

```
SP3(config-if)#ip address 192.168.45.1 255.255.255.0
SP3(config-if)#no shutdown
```

5. Ping de la liaison PE-CE (table classique)

```
SIEGE#ping 192.168.12.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/10/12 ms
```

6. Ping avec référence à la VRF

```
SP1#ping vrf Client_A 192.168.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 16/19/20 ms
```

```
SP3#ping vrf Client_A 192.168.45.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 16/19/20 ms
SP3#
```

Étape 2 : EIGRP sur les routeurs PE

1. Configurer EIGRP AS 1 avec une address-family VRF

SP1

```
SP1#ping 192.168.12.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
SP1#write
Building configuration...
[OK]
SP1#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
SP1(config)#router eigrp 1
SP1(config-router)#address-family ipv4 vrf Client_A
SP1(config-router-af)#network 192.168.12.0 0.0.0.255
SP1(config-router-af)#no auto-summary
SP1(config-router-af)#autonomous-system 100
SP1(config-router-af)#ei
*Oct  9 13:54:44.635: %DUAL-5-NBRCHANGE: EIGRP-IPv4 100: Neighbor 192.168.12.2 (
Ethernet3/0) is up: new adjacencyt
                    ^
% Invalid input detected at '^' marker.

SP1(config-router-af)#exit-address-family
SP1(config-router)#
```

SP3

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 16/19/20 ms
SP3#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
SP3(config)#router eigrp 1
SP3(config-router)#address-family ipv4 vrf Client_A
SP3(config-router-af)#network 192.168.45.0 0.0.0.255
SP3(config-router-af)#no auto-summary
SP3(config-router-af)#autonomous-system 100
SP3(config-router-af)#exi
*Oct  9 13:56:25.307: %DUAL-5-NBRCHANGE: EIGRP-IPv4 100: Neighbor 192.168.45.2
Ethernet3/1) is up: new adjacencyt-
SP3(config-router)#exit-address-family
```

2. Table de routage classique

```
SP1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
C        1.1.1.1 is directly connected, Loopback0
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/11] via 172.16.12.2, 00:52:29, Ethernet3/1
      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/21] via 172.16.12.2, 00:33:19, Ethernet3/1
      172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
C        172.16.12.0/30 is directly connected, Ethernet3/1
L        172.16.12.1/32 is directly connected, Ethernet3/1
O        172.16.23.0/30 [110/20] via 172.16.12.2, 00:51:59, Ethernet3/1
SP1#
```

3. Table de routage de la VRF

```
SP1#sh ip route vrf Client_A

Routing Table: Client_A
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
D       10.10.10.0 [90/307200] via 192.168.12.2, 00:05:39, Ethernet3/0
     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected, Ethernet3/0
L       192.168.12.1/32 is directly connected, Ethernet3/0
SP1#
```

```
SP3#sh ip route vrf Client_A

Routing Table: Client_A
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

D     192.168.1.0/24 [90/307200] via 192.168.45.2, 00:04:47, Ethernet3/1
      192.168.45.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.45.0/24 is directly connected, Ethernet3/1
L        192.168.45.1/32 is directly connected, Ethernet3/1
```

✍ **Y a-t-il des routes EIGRP (D) ?**

Oui. SP1 a une route D vers le LAN SIEGE (10.10.10.0/24) et SP3 a une route D vers le LAN AGENCE (192.168.1.0/24), apprises de leur CE.

✍ **SIEGE et SP1 ont-ils des routes vers les réseaux du client côté AGENCE ?**

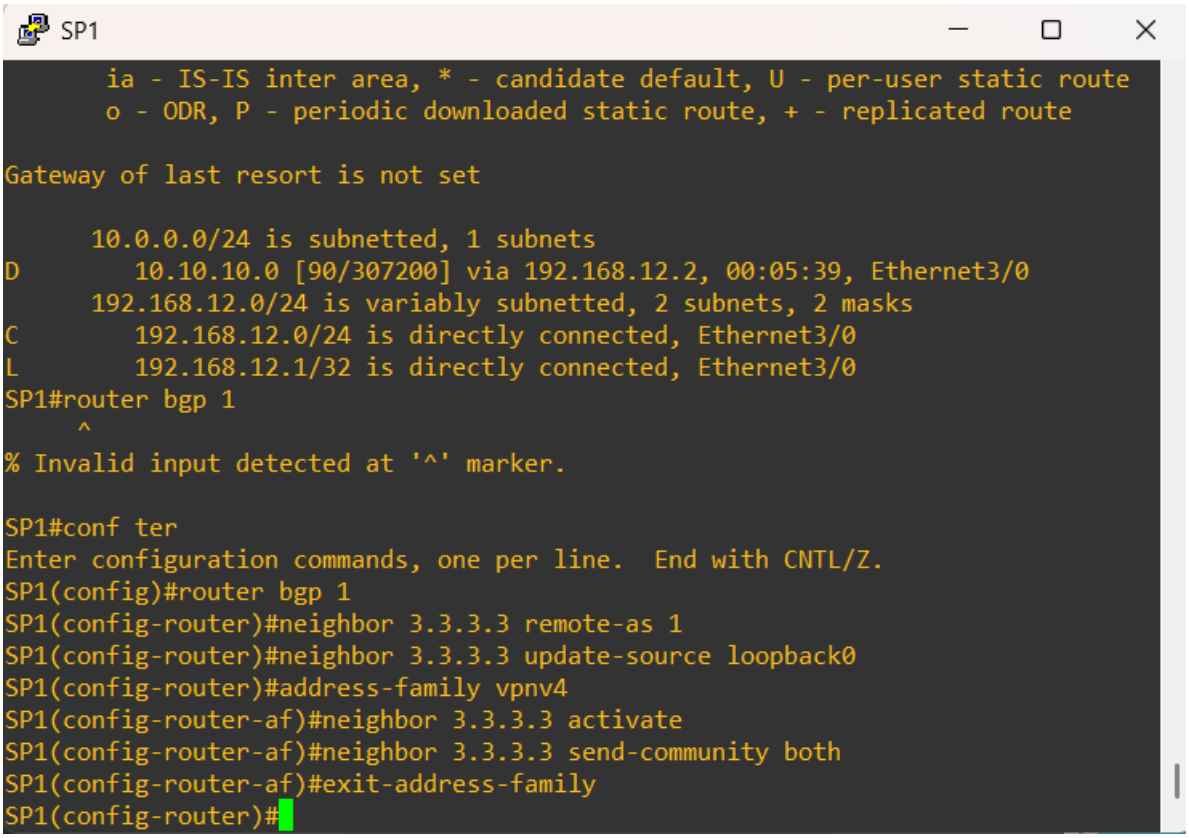
Non. Rien n'est encore échangé entre les deux PE.

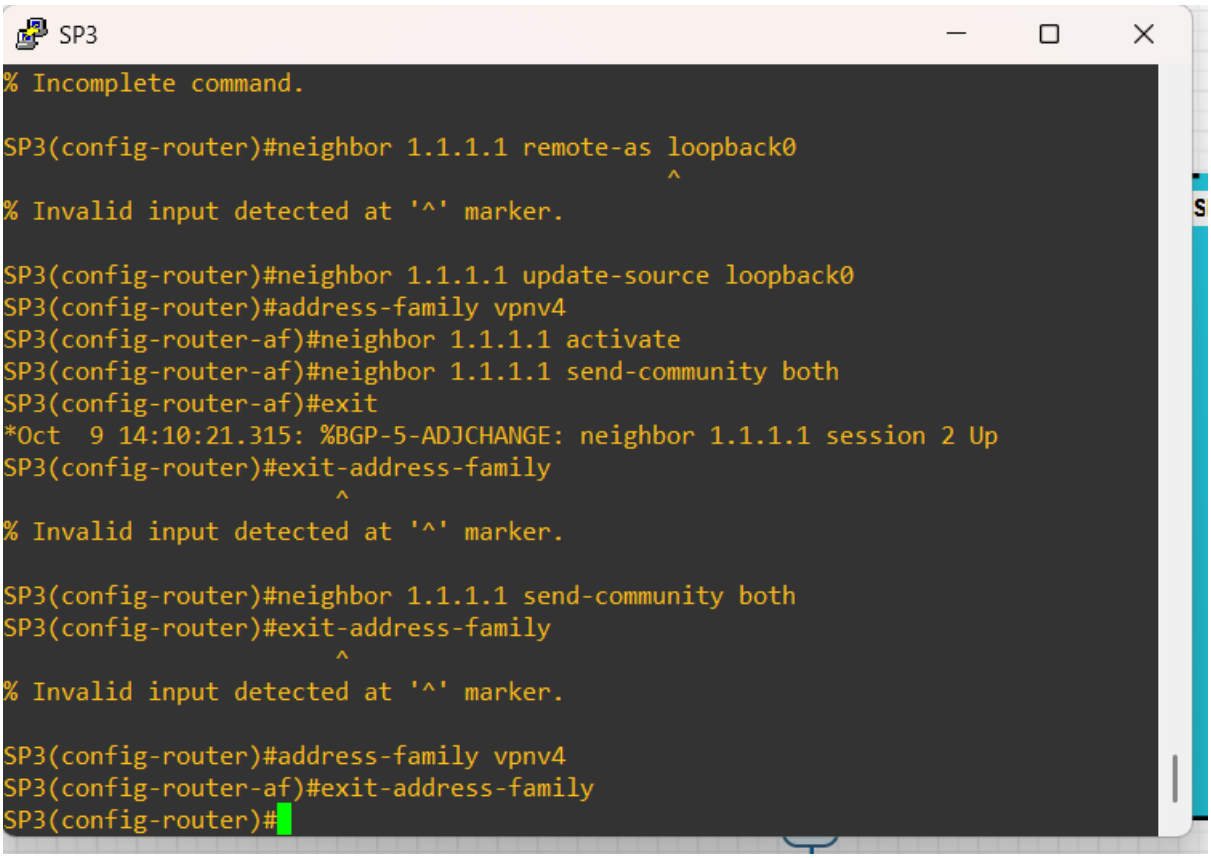
✍ **SP3 et AGENCE ont-ils des routes vers les réseaux du client côté SIEGE ?**

Non, pour la même raison.

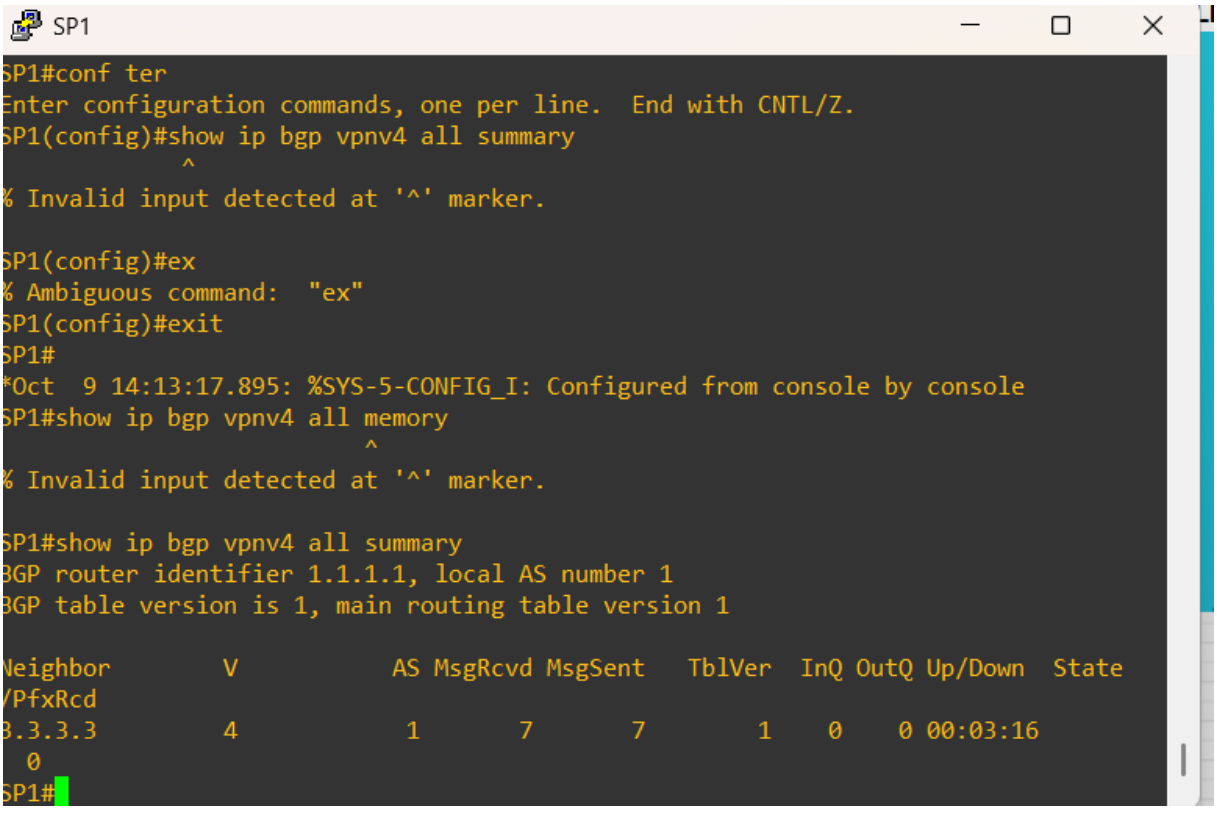
Étape 3 : BGP entre SP1 et SP3 et redistribution

1. Configuration de BGP (VPNv4)





Vérification :



2. Redistribution entre BGP et EIGRP (sur SP1 et SP3) :

```
SP1#show ip bgp vpnv4 all summary
BGP router identifier 1.1.1.1, local AS number 1
BGP table version is 1, main routing table version 1

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State
/PfxRcd
3.3.3.3         4            1       7       7        1    0    0 00:03:16
  0
SP1#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
SP1(config)#router eigrp 1
SP1(config-router)#address-family ipv4 vrf Client_A
SP1(config-router-af)#redistributr bgp 1 metric 1500 4000 200 10 1500
                                   ^
% Invalid input detected at '^' marker.

SP1(config-router-af)#redistribute bgp 1 metric 1500 4000 200 10 1500
SP1(config-router-af)#exit-address-family
SP1(config-router)#router bgp 1
SP1(config-router)#address-family ipv4 vrf Client_A
SP1(config-router-af)#redistribute eigrp 100
SP1(config-router-af)#exit-address-family
SP1(config-router)#
```

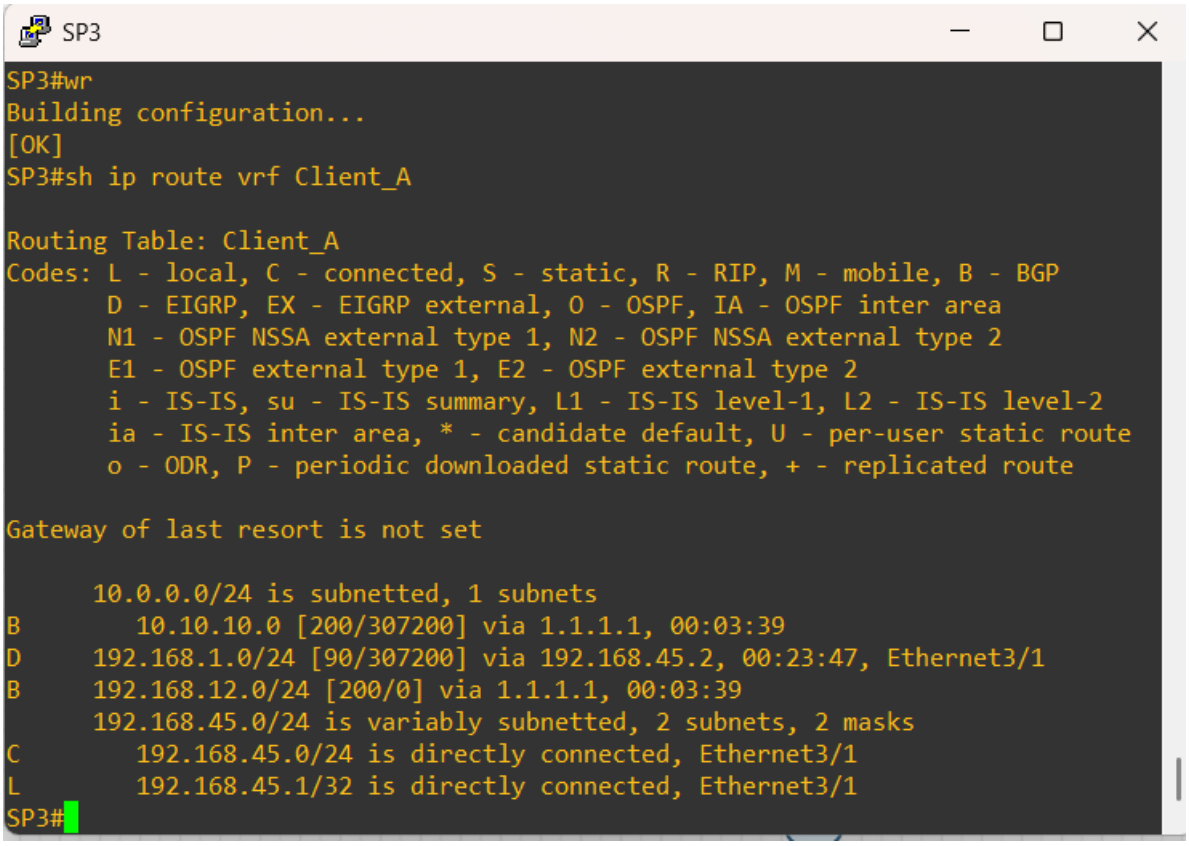
SP3

```
                     ^
% Invalid input detected at '^' marker.

SP3(config-router)#neighbor 1.1.1.1 send-community both
SP3(config-router)#exit-address-family
                     ^
% Invalid input detected at '^' marker.

SP3(config-router)#address-family vpnv4
SP3(config-router-af)#exit-address-family
SP3(config-router)#end
SP3#
*Oct  9 14:17:22.435: %SYS-5-CONFIG_I: Configured from console by consoleconf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SP3(config)#router eigrp 1
SP3(config-router)#address-family ipv4 vrf Client_A
SP3(config-router-af)#redistribute bgp 1 metric 1500 4000 200 10 1500
SP3(config-router-af)#exit-address-family
SP3(config-router)#router bgp 1
SP3(config-router)#address-family ipv4 vrf Client_A
SP3(config-router-af)#redistribute eigrp 100
SP3(config-router-af)#exit-address-family
SP3(config-router)#
```

Vérification :



Étape 4 : tests de connectivité

1. Capture Wireshark entre SP2 et SP3 (le WireShark n'a pas marché)

Connectivité entre le pc windows et mon lab EVE-Ng

```
C:\Users\MSI>ping 192.168.29.128

Envoi d'une requête 'Ping'  192.168.29.128 avec 32 octets de données :
Réponse de 192.168.29.128 : octets=32 temps<1ms TTL=64
Réponse de 192.168.29.128 : octets=32 temps<1ms TTL=64
Réponse de 192.168.29.128 : octets=32 temps<1ms TTL=64
```

Port 22 ouvert(SSH)

```
PS C:\Users\MSI> Test-NetConnection -ComputerName 192.168.29.128 -Port 22


ComputerName     : 192.168.29.128
RemoteAddress    : 192.168.29.128
RemotePort       : 22
InterfaceAlias   : VMware Network Adapter VMnet8
SourceAddress    : 192.168.29.1
TcpTestSucceeded : True
```

2. Pings

```
SIEGE#ping 192.168.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 44/49/56 ms
SIEGE#
```

```
AGENCE#ping 10.10.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/45/60 ms
AGENCE#
```